

```

//create data in the form of "simpleheat" input
var data=[];
for(var i=0;i<dataset.data.length;i++)
{
  data.push(dataset.data[i].screenPos);
}
//Hide the Video Overlay
get('overlay').style.display='none';
//Display The Heat Map
get('canvas').style.display='block';
generateHeatMap(data);
}

```

STEP 2.

Define the generateHeatMap function that renders the heat map and hides the video overlay

```

function generateHeatMap(data){
  /*A heat map generator using "simpleheat.js" (https://
  github.com/mourner/simpleheat) to show your most
  viewed areas in a heatmap representation*/
  window.requestAnimationFrame=
  window.requestAnimationFrame|| window.
  mozRequestAnimationFrame||
  window.webkitRequestAnimationFrame
  || window.msRequestAnimationFrame;

  var heat = simpleheat('canvas').data(data).max(18),
  frame;
  function draw() {
    console.time('draw');
    heat.draw();
    console.timeEnd('draw');
    frame = null;
  }
  draw();
}

```

Inferences:

- The tracking modules are a bit difficult to work with, clmtracker was the fastest of the lot, while trackingsjs was relatively slower and less accurate.
- The different regression models tend to learn at nearly the same pace with the threaded Ridge model being the fastest (as data training was on a separate thread).
- Even with 2-3 hrs of training we were unable to achieve less than 500 pixel accuracy of gaze prediction.

Conclusion:

Computer vision is a rapidly expanding field in the science of computing with amazing potential.

WebGazer seems to be a unique way to leverage it in an attempt to better understand user interaction and interest, that can in turn help us to further enhance user engagement as well as interaction.

The possibilities for WebGazer seem to be endless, from better understanding user choices and interest on an eCommerce website to gaze based interaction in gaming. Even in mobile applications, WebGazer represents an easy way for developers to harness a new way of user interaction. It can prove to be as disruptive in its innovation as VirtualReality and even more widespread because of its simplicity. However the ethical concerns of allowing an application to access your webcam to 'see' you and your surroundings are significant. Such permissions would only be reasonable for completely trustworthy websites or apps and even then would be hard to acquire.

Although both Computer Vision and machine learning are pre-existing technologies, this library is probably the first attempt to combine them in such a novel way. But above all it is the simplicity and ease of use of this javascript library that makes it a truly unique and democratized tool.

With the advent of dedicated vision processing units, this development might lead to interesting possibilities in the fields that we discussed earlier. A computer that can actually see, might be able to do a lot more with the ability to estimate your gaze. From extremely realistic human-machine interaction, to machines that can actually tell when you are having trouble focussing on your work, the machines of the future can and will truly look back at you.>

*Nodes of Interest



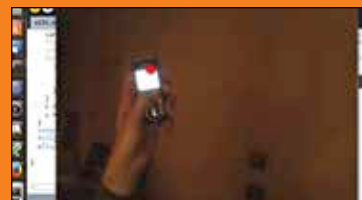
*"Coursera - IBM \$2.4 million funds for startups

>> 20 entrepreneurs to receive \$120,000 in IBM Cloud credits
<http://dgit.in/IBMCrsra>



*"JavaScript from Beginner to Expert

>> 30 day course from Udemy to become front end web developers.
<http://dgit.in/JSUdemy>



*"Light tracking and Air drawing

>> Tutorial for object tracking and using light to draw based on webcam.
<http://dgit.in/AirDraw>